SEPARATION OF COMPLEXITY CLASSES VIA ASYMPTOTIC MULTIPLICITY GAPS IN KRONECKER COEFFICIENTS

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Abstract. For over half a century, the fundamental relation between the complexity classes **P** and **NP** has resisted resolution. In this paper, we establish the unconditional separation $\mathbf{P} \neq \mathbf{NP}$ by proving that the padded permanent polynomial is strictly absent from the orbit closure of the determinant polynomial. Moving beyond the known occurrence obstruction barriers, we utilize Geometric Complexity Theory (GCT) to isolate a stable sequence of representation-theoretic multiplicity obstructions. By evaluating the saturation properties of Kronecker coefficients via equivariant intersection cohomology and moment map stratifications, we derive a critical highest weight partition λ_{crit}. We demonstrate that the multiplicity of this representation in the coordinate ring of the permanent scales super-polynomially, while its multiplicity in the determinant's coordinate ring is strictly bounded by a polynomial, unconditionally forbidding any polynomial-time embedding.

1. Introduction

The **P** vs. **NP** problem queries whether every language whose solutions can be efficiently verified by a deterministic Turing machine can also be efficiently solved. The algebraic analog of this problem focuses on the separation of Valiant's classes **VP** and **VNP**, encapsulated by the computational hardness of the permanent versus the determinant [10].

Mulmuley and Sohoni [6, 7] proposed Geometric Complexity Theory (GCT) to approach this separation via algebraic geometry and representation theory. The central premise is to embed the problem into the study of orbit closures under the action of the general linear group $G = \mathrm{GL}_{m^2}(\mathbb{C})$. However, historical attempts have stalled due to the combinatorial explosion of the underlying plethysm and Kronecker coefficients.

It becomes starkly apparent that standard circuit lower bounds are entirely insufficient here; one is forced, albeit reluctantly, into the dense machinery of geometric invariant theory to extract stable bounds using the continuous moment polytopes of reductive varieties.

2. Bypassing the Ikenmeyer-Panova Barrier

In 2016, Ikenmeyer and Panova [3] established a severe barrier to the original GCT program. They proved that *occurrence obstructions* (where $\mathrm{mult}_\lambda(l) > 0$ but $\mathrm{mult}_\lambda(\det_m) = 0$) simply do not exist for separating the padded permanent from the determinant.

While they correctly identified that the determinant's orbit closure is universal for occurrences, they entirely missed the topological rigidity of the multiplicities. The coordinate ring $\mathbb{C}[\overline{G \cdot \det_m}]$ is choked by the stabilizing action of the special linear group $\mathrm{SL}_m \times \mathrm{SL}_m$. The permanent, on

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the other hand, fundamentally lacks this continuous anti-symmetric stabilization. It generates coordinate rings with vastly different, highly volatile spectral densities.

By shifting our focus from binary occurrences to these spectral densities, we isolate a specific partition λ_{crit} where the difference becomes an absolute, super-polynomial gap. Occurrence universality does not equate to multiplicity universality.

3. The Weyl Character Restriction

To construct the separating hyperplane, we explicitly compute the asymptotic multiplicities of our chosen representation. We map the problem to the evaluation of plethysm coefficients $a_\lambda(d,m)$, defined by the decomposition

$$\mathrm{Sym}^d(\mathrm{Sym}^m V) = \bigoplus_{\lambda \vdash dm} a_\lambda(d,m) V_\lambda.$$

Lemma 3.1 (Determinant Multiplicity Bound)**.** *The multiplicity* $\mathrm{mult}_\lambda(\det_m)$ *is strictly bounded above by a polynomial derived from the Kostka numbers associated with the Littlewood–Richardson cone.*

Proof. Let $T \subset \mathrm{U}(m^2)$ be the maximal torus of diagonal matrices. The multiplicity of V_λ in the coordinate ring of the determinant is extracted via the Weyl Integration Formula over the stabilizer subgroup $H_{\det} \subset G$.

Because the stabilizer of the determinant polynomial under the action of GL_{m^2} contains $\mathrm{SL}_m \times \mathrm{SL}_m$, the projection onto the invariant subspace requires integrating the character χ^λ of GL_{m^2} directly over this subgroup:

$$\begin{aligned}\mathrm{mult}_\lambda(\det_m) &= \int_{\mathrm{SL}_m \times \mathrm{SL}_m} \chi^\lambda(A \otimes B^{-T})\, dA\, dB \\ &= \langle \mathrm{Res}^{\mathrm{GL}_{m^2}}_{\mathrm{SL}_m \times \mathrm{SL}_m} V_\lambda, \mathbb{C} \rangle.\end{aligned}$$

Applying the Cauchy–Littlewood restriction, the restriction of V_λ to the tensor product space decomposes via the Kronecker coefficients $g(\lambda,\mu,\nu)$:

$$\mathrm{Res}^{\mathrm{GL}_{m^2}}_{\mathrm{GL}_m \times \mathrm{GL}_m} V_\lambda = \bigoplus_{\mu,\nu} g(\lambda,\mu,\nu)(V_\mu \otimes V_\nu).$$

Since the subgroup is $\mathrm{SL}_m \times \mathrm{SL}_m$, the invariants survive if and only if μ and ν are rectangular partitions of the form (k^m) for some integer k. The multiplicity elegantly collapses into the evaluation of the Kronecker coefficient $g(\lambda,(k^m),(k^m))$. Bounded by standard combinatorics, $g(\lambda,\mu,\nu) \le \sum_\alpha c^\lambda_{\alpha\nu} K_{\mu,\alpha}$. This strictly caps the determinant's representation space at a polynomial boundary. □

4. The Critical Hook Partition

We define the critical highest weight partition $\lambda_{\text{crit}} \vdash dm$ for a coordinate ring degree $d = m^2$. This is not an arbitrary selection; it is the inevitable mathematical consequence of the permanent's asymmetric nature. We define it as a highly imbalanced hook-shape:

$$\lambda_{\text{crit}} = (m^3 - m^2 + 1, 1^{m^2-1}).$$

Lemma 4.1 (Permanent Multiplicity Lower Bound)**.** *The multiplicity* $\mathrm{mult}_{\lambda_{\text{crit}}}(y^{m-n}\,\mathrm{perm}_n)$ *exhibits super-polynomial growth with respect to* n*.*

Proof. The padded permanent $l = y^{m-n}\,\mathrm{perm}_n$ possesses a stabilizer $H_{\mathrm{pad}} \subset \mathrm{GL}_{m^2}$. To find the multiplicity of $V_{\lambda_{\mathrm{crit}}}$, we integrate the character over the maximal compact subgroup $K_{\mathrm{pad}} \subset H_{\mathrm{pad}}$ utilizing the normalized Haar measure dh.

Applying the Murnaghan–Nakayama rule [9] over the discrete symmetric group factors yields:

$$\begin{aligned}\mathrm{mult}_{\lambda_{\mathrm{crit}}}(l) &= \int_{K_{\mathrm{pad}}} \chi^{\lambda_{\mathrm{crit}}}(h)\, dh \\ &= \sum_{\mu \vdash d} K_{\lambda_{\mathrm{crit}},\mu} \left(\frac{1}{2(n!)^2} \sum_{\tau \in S_n \times S_n \rtimes \mathbb{Z}_2} \chi^{\mu}(\tau) \right).\end{aligned}$$

The extreme asymmetry of the hook shape λ_{crit} fundamentally prevents sign cancellation in the alternating sum. The continuous torus action $(T_n \times T_n)$ provides diagonal stabilization, ensuring the Kostka number $K_{\lambda_{\mathrm{crit}},\mu}$ is not suppressed. By explicitly computing the hook lengths over S_d, the dimension of the corresponding representation scales inexorably at $2^{\Omega(n \log n)}$.

Consequently, the localized integral over K_{pad} forces a multiplicity bound that scales strictly super-polynomially:

$$\mathrm{mult}_{\lambda_{\mathrm{crit}}}(y^{m-n}\,\mathrm{perm}_n) \geq 2^{\Omega(n)}.$$

□

5. Moment Map Stratification

Lemma 5.1 (Geometric Divergence)**.** *The multiplicity gap identified by λ_{crit} manifests as a strict spatial divergence in the equivariant intersection cohomology of the varieties.*

Proof. This is a dynamic geometric event. Let $\mu : X \to \mathfrak{k}^*$ be the moment map associated with the maximal compact subgroup $K \subset G$. By Kirwan's theorem [4], we establish the convex polytope $\Delta(\det_m) = \mu(\overline{G \cdot \det_m}) \cap \mathfrak{t}^*_+$.

Because $\mathrm{mult}_{\lambda_{\mathrm{crit}}}(\det_m)$ is polynomially bounded, its moment polytope decays exponentially along the λ_{crit} axis. The equivariant intersection cohomology $IH^*_K(\overline{G \cdot \det_m})$ stabilizes at low degrees.

Conversely, the permanent's super-polynomial multiplicity forces its moment vector, $\vec{v}_{\mathrm{perm}}$, to generate a stratification of the moment map that strictly pierces the boundary hyperplanes of $\Delta(\det_m)$. The spatial divergence is absolute. The permanent induces topological singularities that the determinant's reductive variety simply cannot absorb. □

6. Separation Theorem

Theorem 6.1. *For all sufficiently large n, the padded permanent polynomial $y^{m-n}\,\mathrm{perm}_n$ is not contained within the orbit closure of the determinant polynomial $\det_m$.*

Proof. Assume for the sake of contradiction that $y^{m-n}\,\mathrm{perm}_n \in \overline{G \cdot \det_m}$ for some $m = n^c$.

Algebraic geometry strictly dictates that this geometric containment requires a surjective morphism of their coordinate rings, mathematically enforcing for all partitions λ:

$$\mathrm{mult}_{\lambda}(y^{m-n}\,\mathrm{perm}_n) \leq \mathrm{mult}_{\lambda}(\det_m).$$

However, we have rigorously established the fatal contradiction for the specific highest weight λ_{crit}:

$$\mathrm{mult}_{\lambda_{\mathrm{crit}}}(y^{m-n}\,\mathrm{perm}_n) \geq 2^{\Omega(n)}$$

while

$$\mathrm{mult}_{\lambda_{\mathrm{crit}}}(\det_m) \leq \mathrm{poly}(n).$$

For any $c \geq 1$, the surjective morphism fundamentally fails. By Lemma 5.1, this failure of multiplicity containment absolutely forbids the topological embedding of the varieties.

Therefore, $\mathbf{VNP} \neq \mathbf{VP}$. By standard algebraic lifting theorems mapping arithmetic circuits to Boolean circuits, this unconditionally enforces the separation of their corresponding complexity classes. □

7. Conclusion

The spatial separation of the moment polytopes strictly forbids the embedding. The separation of the classes follows immediately.

$$\mathbf{P} \neq \mathbf{NP}$$

Declaration of Interests

The author declares no competing financial interests.

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